



LAPORAN UJIAN TENGAH SEMESTER

Pemrograman Web / Framework

Sistem Manajemen Perpustakaan Jaya

Nama	: Bagas Syahputra
NIM	: 09010582529101
Mata Kuliah	: Pemrograman Web
Framework	: Laravel 11
Database	: SQLite

Oktober 2026

A. Source Code Aplikasi Laravel

Struktur Folder Proyek

Struktur Direktori

```
uts_bagas_09010582529101/
├── app/
│   ├── Http/Controllers/
│   │   ├── AuthController.php
│   │   └── BookController.php
│   └── Models/
│       ├── Book.php
│       └── Category.php
├── database/
│   ├── migrations/
│   │   ├── create_categories_table.php
│   │   └── create_books_table.php
│   └── seeders/
│       ├── DatabaseSeeder.php
│       ├── UserSeeder.php
│       ├── CategorySeeder.php
│       └── BookSeeder.php
├── resources/views/
│   ├── layouts/app.blade.php
│   ├── auth/login.blade.php
│   └── books/
│       ├── index.blade.php
│       ├── create.blade.php
│       ├── edit.blade.php
│       └── show.blade.php
└── routes/web.php
```

Routes — routes/web.php

routes/web.php

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\AuthController;
use App\Http\Controllers\BookController;

Route::get('/', function () {
    return redirect('/books');
});

// Authentication
Route::get('/login', [AuthController::class, 'showLogin'])->name('login');
Route::post('/login', [AuthController::class, 'login']);
Route::post('/logout', [AuthController::class, 'logout'])->name('logout');

// Book CRUD (harus login)
Route::middleware('auth')->group(function () {
    Route::resource('books', BookController::class);
});
```

Model — app/Models/Category.php

app/Models/Category.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\HasMany;

class Category extends Model
{
    use HasFactory;

    protected $fillable = ['name', 'description'];

    public function books(): HasMany
    {
        return $this->hasMany(Book::class);
    }
}
```

Model — app/Models/Book.php

app/Models/Book.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class Book extends Model
{
    use HasFactory;

    protected $fillable = [
        'category_id', 'title', 'author',
        'publisher', 'year', 'stock',
    ];

    public function category(): BelongsTo
    {
        return $this->belongsTo(Category::class);
    }
}
```

Controller — AuthController.php

app/Http/Controllers/AuthController.php

```
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;

class AuthController extends Controller
{
    public function showLogin()
    {
        return view('auth.login');
    }

    public function login(Request $request)
    {
        $credentials = $request->validate([
            'email' => 'required|email',
            'password' => 'required',
        ]);

        if (Auth::attempt($credentials)) {
            $request->session()->regenerate();
            return redirect()->intended('/books')->with('success', 'Berhasil login!');
        }

        return back()->withErrors([
            'email' => 'Email atau password salah.',
        ])->onlyInput('email');
    }

    public function logout(Request $request)
    {
        Auth::logout();
        $request->session()->invalidate();
        $request->session()->regenerateToken();
        return redirect('/login')->with('success', 'Berhasil logout!');
    }
}
```

Controller — BookController.php

app/Http/Controllers/BookController.php

```

<?php

namespace App\Http\Controllers;

use App\Models\Book;
use App\Models\Category;
use Illuminate\Http\Request;

class BookController extends Controller
{
    public function index(Request $request)
    {
        $query = Book::with('category');

        if ($request->filled('search')) {
            $search = $request->search;
            $query->where(function ($q) use ($search) {
                $q->where('title', 'like', '%' . $search . '%')
                    ->orWhere('author', 'like', '%' . $search . '%');
            });
        }

        if ($request->filled('category_id')) {
            $query->where('category_id', $request->category_id);
        }

        $books = $query->latest()->paginate(10);
        $categories = Category::all();
        return view('books.index', compact('books', 'categories'));
    }

    public function create()
    {
        $categories = Category::all();
        return view('books.create', compact('categories'));
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'title' => 'required|string|max:255',
            'author' => 'required|string|max:255',
            'publisher' => 'required|string|max:255',
            'year' => 'required|integer|min:1900|max:' . date('Y'),
            'stock' => 'required|integer|min:0',
            'category_id' => 'required|exists:categories,id',
        ]);

        Book::create($validated);
        return redirect()->route('books.index')->with('success', 'Buku berhasil ditambah');
    }
}

```

```
public function show(Book $book)
{
    $book→load('category');
    return view('books.show', compact('book'));
}

public function edit(Book $book)
{
    $categories = Category::all();
    return view('books.edit', compact('book', 'categories'));
}

public function update(Request $request, Book $book)
{
    $validated = $request→validate([
        'title' ⇒ 'required|string|max:255',
        'author' ⇒ 'required|string|max:255',
        'publisher' ⇒ 'required|string|max:255',
        'year' ⇒ 'required|integer|min:1900|max:' . date('Y'),
        'stock' ⇒ 'required|integer|min:0',
        'category_id' ⇒ 'required|exists:categories,id',
    ]);

    $book→update($validated);
    return redirect()→route('books.index')→with('success', 'Buku berhasil diperbarui');
}

public function destroy(Book $book)
{
    $book→delete();
    return redirect()→route('books.index')→with('success', 'Buku berhasil dihapus');
}
}
```

B. Migration dan Seeder

Migration — create_categories_table

database/migrations/2024_01_01_000001_create_categories_table.php

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('categories', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->text('description')->nullable();
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('categories');
    }
};
```

Migration — create_books_table

```
database/migrations/2024_01_01_000002_create_books_table.php

<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('books', function (Blueprint $table) {
            $table->id();
            $table->foreignId('category_id')->constrained('categories')->onDelete('cascade');
            $table->string('title');
            $table->string('author');
            $table->string('publisher');
            $table->integer('year');
            $table->integer('stock')->default(0);
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('books');
    }
};
```

Struktur Database

Tabel	Field	Tipe	Keterangan
categories	id	BIGINT	Primary Key
categories	name	VARCHAR	Nama kategori
categories	description	TEXT	Deskripsi kategori
categories	timestamps	TIMESTAMP	Waktu pembuatan/perubahan
books	id	BIGINT	Primary Key
books	category_id	BIGINT	Foreign Key ke categories

Tabel	Field	Tipe	Keterangan
books	title	VARCHAR	Judul buku
books	author	VARCHAR	Penulis
books	publisher	VARCHAR	Penerbit
books	year	INTEGER	Tahun terbit
books	stock	INTEGER	Jumlah stok
books	timestamps	TIMESTAMP	Waktu pembuatan/perubahan

Seeder — UserSeeder.php

database/seeder/UserSeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
use App\Models\User;
use Illuminate\Support\Facades\Hash;

class UserSeeder extends Seeder
{
    public function run(): void
    {
        User::create([
            'name' => 'Bagas Syahputra',
            'email' => 'bagas@jayapustaka.com',
            'password' => Hash::make('bagas123'),
        ]);
    }
}
```

Seeder — CategorySeeder.php

database/seeder/CategorySeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
use App\Models\Category;

class CategorySeeder extends Seeder
{
    public function run(): void
    {
        $categories = [
            ['name' => 'Fiksi', 'description' => 'Buku cerita fiksi, novel, dan karya sastra'],
            ['name' => 'Non-Fiksi', 'description' => 'Buku berbasis fakta, biografi, dan dokumentasi'],
            ['name' => 'Teknologi', 'description' => 'Buku tentang teknologi informasi dan komunikasi'],
            ['name' => 'Sains', 'description' => 'Buku ilmu pengetahuan alam, fisika, kimia, dan biologi'],
        ];

        foreach ($categories as $category) {
            Category::create($category);
        }
    }
}
```

Seeder — BookSeeder.php

database/seeder/BookSeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
use App\Models\Book;

class BookSeeder extends Seeder
{
    public function run(): void
    {
        $books = [
            ['category_id'⇒3, 'title'⇒'Pemrograman Web dengan PHP', 'author'⇒'Lukman', 'publisher'⇒'Elex Media Komputindo'],
            ['category_id'⇒1, 'title'⇒'Negeri 5 Menara', 'author'⇒'Ahmad Fuadi', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒4, 'title'⇒'Kosmos', 'author'⇒'Carl Sagan', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒2, 'title'⇒'Sapiens', 'author'⇒'Yuval Noah Harari', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒1, 'title'⇒'Perahu Kertas', 'author'⇒'Dee Lestari', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒3, 'title'⇒'Belajar Laravel untuk Pemula', 'author'⇒'Muhammad Nur Hafidza', 'publisher'⇒'Elex Media Komputindo'],
            ['category_id'⇒4, 'title'⇒'Fisika Dasar', 'author'⇒'Halliday & Resnick', 'publisher'⇒'Erlangga'],
            ['category_id'⇒2, 'title'⇒'Atomic Habits', 'author'⇒'James Clear', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒1, 'title'⇒'Dilan 1990', 'author'⇒'Pidi Baiq', 'publisher'⇒'Gramedia Pustaka Utama'],
            ['category_id'⇒4, 'title'⇒'Pengantar Kimia Organik', 'author'⇒'Fessenden', 'publisher'⇒'Erlangga'],
        ];

        foreach ($books as $book) {
            Book::create($book);
        }
    }
}
```

Seeder — DatabaseSeeder.php

database/seeder/DatabaseSeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    use WithoutModelEvents;

    public function run(): void
    {
        $this->call([
            UserSeeder::class,
            CategorySeeder::class,
            BookSeeder::class,
        ]);
    }
}
```

C. Screenshot Halaman Aplikasi

Keterangan: Berikut adalah hasil tangkapan layar (screenshot) dari setiap antarmuka aplikasi Perpustakaan Jaya yang telah berjalan sempurna pada lingkungan lokal dengan akun pengguna **Bagas Syahputra** (bagas@jayapustaka.com).

1. Halaman Login

Tampilan halaman form login sistem dengan judul **Perpustakaan Jaya** dan akun autentikasi pengguna.



The screenshot shows a login page for 'Perpustakaan Jaya'. The page has a dark blue header with the title 'Perpustakaan Jaya' and the subtitle 'Silakan masuk ke sistem'. Below the header is a white login form with fields for 'Email' and 'Password', a 'Masuk' button, and a footer with the text 'Akun: bagas@jayapustaka.com / bagas123'.

Perpustakaan Jaya
Silakan masuk ke sistem

Email
Email anda

Password
Password anda

Masuk

Akun: bagas@jayapustaka.com / bagas123

© 2026 Perpustakaan Jaya — Sistem Informasi Manajemen Buku

2. Dashboard / Daftar Buku

Tampilan dashboard utama menampilkan ringkasan data (Total Buku: 10, Kategori: 4, Total Stok: 127), filter pencarian & kategori, tabel 10 koleksi buku, dan profil pengguna aktif (Bagas Syahputra).

Perpustakaan Jaya

Bagas SyahputraKeluar

Beranda / Daftar Buku

Total Buku
10

Kategori
4

Total Stok
127

Data Buku

Tambah Buku

Cari judul / penulis...

Semua Kategori

Cari

No	Judul	Penulis	Penerbit	Tahun	Kategori	Stok	Aksi
1	Pemrograman Web dengan PHP	Lukmanul Hakim	Lokomedia	2014	Teknologi	9	  
2	Negeri 5 Menara	Ahmad Fuadi	Gramedia Pustaka Utama	2009	Fiksi	18	  
3	Kosmos	Carl Sagan	Random House	1980	Sains	5	  
4	Sapiens: Riwayat Singkat Umat Manusia	Yuval Noah Harari	Pustaka Alvabet	2017	Non-Fiksi	14	  
5	Perahu Kertas	Dee Lestari	Bentang Pustaka	2009	Fiksi	7	  
6	Belajar Laravel untuk Pemula	Muhamad Nauval Azhar	Leanpub	2021	Teknologi	22	  
7	Fisika Dasar	Halliday & Resnick	Erlangga	2010	Sains	11	  
8	Atomic Habits	James Clear	Gramedia Pustaka Utama	2019	Non-Fiksi	25	  
9	Dilan 1990	Pidi Baiq	Pastel Books	2014	Fiksi	13	  
10	Pengantar Kimia Organik	Fessenden & Fessenden	Erlangga	2006	Sains	3	  

© 2026 Perpustakaan Jaya — Sistem Informasi Manajemen Buku

3. Halaman Tambah Buku

Tampilan form input data buku baru yang dilengkapi input judul, penulis, penerbit, tahun terbit, stok, serta dropdown relasi kategori.

Perpustakaan Jaya

Bagas Syahputra

Keluar

Beranda / Tambah Buku

Kembali

Tambah Buku Baru

Judul Buku

Masukkan judul buku

Penulis

Nama penulis

Penerbit

Nama penerbit

Tahun Terbit

Contoh: 2024

Stok

0

Kategori

-- Pilih Kategori --

Simpan

Batal

© 2026 Perpustakaan Jaya — Sistem Informasi Manajemen Buku

4. Halaman Edit Buku

Tampilan form edit buku yang otomatis memuat data buku yang dipilih untuk diubah.

Perpustakaan Jaya

Bagas Syahputra

Keluar

Beranda / Edit Buku

Kembali

Edit Buku

Judul Buku

Pemrograman Web dengan PHP

Penulis

Lukmanul Hakim

Penerbit

Lokomedia

Tahun Terbit

2014

Stok

9

Kategori

Teknologi

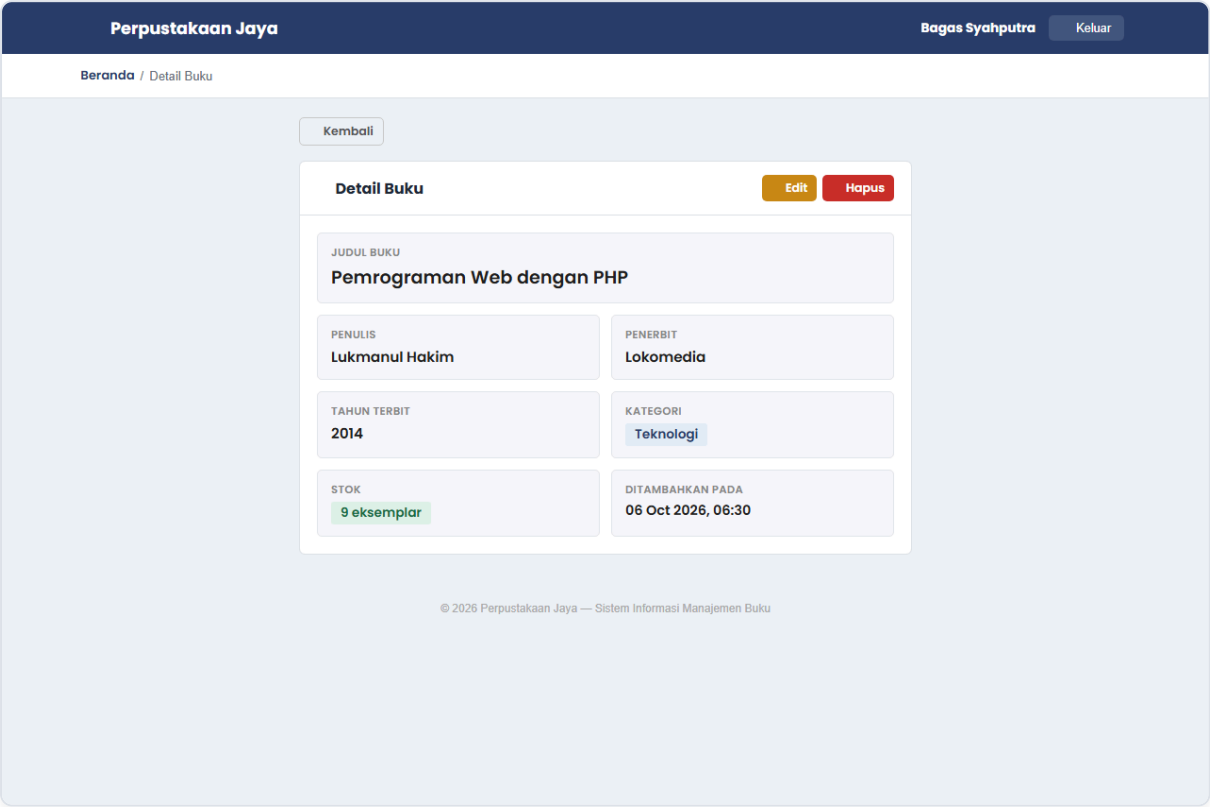
Perbarui

Batal

© 2026 Perpustakaan Jaya — Sistem Informasi Manajemen Buku

5. Halaman Detail Buku

Tampilan halaman rincian detail buku yang menampilkan seluruh atribut buku secara lengkap beserta tombol aksi navigasi.



D. URL Repository GitHub

Source code lengkap aplikasi ini tersedia di repository GitHub berikut:

github.com/pratamabagas748-debug/uts_bagas_09010582529101

Informasi Repository

Keterangan	Nilai
URL	<code>https://github.com/pratamabagas748-debug/uts_bagas_09010582529101</code>
Branch	<code>master</code>
Framework	Laravel 11
Database	SQLite

Cara Menjalankan

Terminal / Command Prompt

```
# Clone repository
git clone https://github.com/pratamabagas748-debug/uts_bagas_09010582529101.git
cd uts_bagas_09010582529101

# Install dependencies
composer install

# Setup environment
copy .env.example .env
php artisan key:generate

# Jalankan migration dan seeder
php artisan migrate:fresh --seed

# Jalankan server
php artisan serve
```

Akun Login

Keterangan	Nilai
URL Aplikasi	http://127.0.0.1:8000
Email	bagas@jayapustaka.com
Password	bagas123